

5.X Formulae Used in the Evaluation of Polyherbal Anti-Dandruff Shampoo

The following formulae were used for the calculation and interpretation of the physicochemical and performance evaluation parameters of the formulated polyherbal anti-dandruff shampoo.

1. Percentage Yield of Herbal Extract

The percentage yield of the dried herbal extract was calculated using the following equation:

$$\text{Percentage Yield (\%)} = \frac{\text{Weight of Dried Extract}}{\text{Weight of Crude Plant Material}} \times 100$$

Reference: [entity\[\["book", "Harborne's Phytochemical Methods", "J. B. Harborne"\]\]](#)¹

2. Percentage Solid Content

The percentage of solid content present in the shampoo formulation was calculated using the following formula:

$$\text{Percentage Solid Content (\%)} = \frac{\text{Weight of Dried Residue}}{\text{Weight of Shampoo Sample}} \times 100$$

Reference: [entity\[\["book", "Harry's Cosmeticology", "Harry's Cosmeticology textbook"\]\]](#)²

3. Foaming Ability

Foaming ability was determined by measuring the increase in foam volume after shaking.

$$\text{Foaming Ability} = \text{Final Foam Volume} - \text{Initial Liquid Volume}$$

Reference: [entity\[\["book", "Harry's Cosmeticology", "Harry's Cosmeticology textbook"\]\]](#)²

4. Foam Stability

Foam stability was calculated by measuring the foam volume after a specified period.

$$\text{Foam Stability (\%)} = \frac{\text{Foam Volume After Specified Time}}{\text{Initial Foam Volume}} \times 100$$

Reference: Harry's Cosmeticology²

5. Detergency Power

The detergency efficiency of the shampoo was calculated using the following equation:

$$\text{Detergency (\%)} = \frac{\text{Weight of Dirt Before Washing} - \text{Weight of Dirt After Washing}}{\text{Weight of Dirt Before Washing}} \times 100$$

Reference: [Entity\["book", "Cosmetic Science and Technology", "B. M. Mithal"\]](#)³

6. Surface Tension (Stalagmometer Method)

Surface tension was determined by the stalagmometer method using the following equation:

$$\gamma_2 = \gamma_1 \times \frac{\rho_2 \times n_1}{\rho_1 \times n_2}$$

Where,

- γ = Surface tension
- ρ = Density
- n = Number of drops

Reference: [Entity\["book", "Physical Pharmacy", "Alfred Martin"\]](#)⁴

7. Viscosity (Ostwald Viscometer Method)

Viscosity was calculated using the Ostwald viscometer equation.

$$\eta_2 = \eta_1 \times \frac{\rho_2 \times t_2}{\rho_1 \times t_1}$$

Where,

- η = Viscosity
- ρ = Density
- t = Flow time

Reference: Alfred Martin, Physical Pharmacy⁴

8. Mean

The arithmetic mean was calculated as follows:

$$\bar{x} = \frac{\sum x}{n}$$

Where,

- x = Individual observation
- n = Number of observations

Reference: [Entity](#)["book", "Fundamentals of Biostatistics", "Bernard Rosner"]⁵

9. Standard Deviation

Standard deviation was calculated using the following formula:

$$SD = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}}$$

Where,

- x = Individual observation
- $\bar{x}$ = Mean
- n = Number of observations

Reference: Fundamentals of Biostatistics⁵

10. Relative Standard Deviation (Optional)

$$RSD(\%) = \frac{SD}{Mean} \times 100$$

Reference: [Entity](#)["book", "ICH Q2(R2) Validation of Analytical Procedures", "International Council for Harmonisation guideline"]⁶

11. Percentage Moisture Loss

The percentage moisture loss during the stability study was calculated as follows:

$$\text{Moisture Loss (\%)} = \frac{\text{Initial Weight} - \text{Final Weight}}{\text{Initial Weight}} \times 100$$

Reference: [Entity](#)["organization", "International Council for Harmonisation", "ICH"]⁷ Stability Guidelines⁷

12. pH Determination

The pH of the shampoo formulation was determined directly using a calibrated digital pH meter. Therefore, no mathematical calculation was required.

Reference: Indian Pharmacopoeia⁸

13. Wetting Time

Wetting time was determined by recording the time (in seconds) required for the canvas disc to sink completely into the shampoo solution. Hence, no mathematical formula was required.

Reference: Harry's Cosmeticology²

14. Dirt Dispersion Test

The dirt dispersion test was performed as a qualitative assessment of cleansing efficiency. No mathematical calculation was required.

Reference: Harry's Cosmeticology²

15. Conditioning Performance

Conditioning performance was evaluated qualitatively based on softness, smoothness, ease of combing, and overall feel after washing. Therefore, no mathematical formula was required.

Reference: Harry's Cosmeticology²

References (Vancouver Style)

1. Harborne JB. *Phytochemical Methods: A Guide to Modern Techniques of Plant Analysis*. 3rd ed. London: Chapman & Hall; 1998.
2. Barel AO, Paye M, Maibach HI, editors. *Harry's Cosmeticology*. 9th ed. New York: Chemical Publishing Co.; 2015.
3. Mithal BM. *A Handbook of Cosmetics*. 2nd ed. New Delhi: Vallabh Prakashan; 2008.
4. Martin A, Bustamante P, Chun AHC. *Physical Pharmacy*. 5th ed. Philadelphia: Lippincott Williams & Wilkins; 2006.
5. Rosner B. *Fundamentals of Biostatistics*. 8th ed. Boston: Cengage Learning; 2016.
6. International Council for Harmonisation. *ICH Q2(R2): Validation of Analytical Procedures*. Geneva: ICH; 2023.

7. International Council for Harmonisation. *ICH Q1A(R2): Stability Testing of New Drug Substances and Products*. Geneva: ICH; 2003.
8. Indian Pharmacopoeia Commission. *Indian Pharmacopoeia*. Ghaziabad: IPC; Latest edition.